



Phase 1 - Research **deevi**

Synthesis

June 2018

Process

Phase 1

Definition of relevant areas to tackle; Running user research; Setting tech requirements; Running initial data analysis

Phase 2

Building & testing of basic UI, functionality & data infrastructure and classification model

Phase 3

Test solution with pilot(s); launch venture

We are here

Phase 1 Goals

*Identify **who** is going to be impacted by our solution, discover **what problems** people have in the factory within the field of quality control, and understand **how** our solution could be integrated in different factory settings.*

HYPOTHESES

- ▶ H0. Our idea addresses a relevant area (Sorting and QA/C). 
- ▶ H1a. Our product idea is solving a real *customer* need. 
- ▶ H1b. The customer wants to invest in a solution like ours because H1a is met. 
- ▶ H2a. We assume that in the *majority* of the cases we need to incorporate our solution into existing solutions and hardware. 
- ▶ H2b. We assume that in the *minority* of the cases we need to incorporate our solution into factories via new hardware bought. 

Agenda

01

**METHODS
& PERSONAS**

02

**EXPERIENCE
MAP**

03

TOP FINDINGS

04

**OPPORTUNITY
AREAS**

05

**DESIGN
PRICIPLES**

06

**OPEN
QUESTIONS**

07

NEXT STEPS

01 Methods & Personas

WHAT WE DID

WE CONDUCTED 8 FIRST-RD INTERVIEWS

with companies located in Germany, Sweden and France from different industries, including food processing, Pharma packaging and automotive.

WE VISITED 2 COMPANIES IN GERMANY

to observe and interview people in the factory context.

User Personas



Diligent Diana



Expert Erik



Line Mgr Leo



Diligent Diana

30-55, Factory Worker for manual assembly
Female

PRECISE

PATIENT

PUNCTUAL

I have a great amount of responsibility: I need to make sure that the boards I am putting together go error-free through my hands. That's 200+ possibilities for fails!

ABOUT

Diana is a 39 year-old mom of two, that works on a three-shift basis in the factory of her company, mainly responsible for manual montage of different parts, at a specific step of the production process.

GOALS

- Putting together as many boards as possible per hour without making any mistakes. *The display above the assembly line which shows (quality) goal error rates motivates her to keep the error score low.*
- Discovering mistakes from previous production steps in parts or the board, makes her feel proud.

FRUSTRATIONS

- Working 8h shifts, especially at night, doing the same repetitive task, with only one big 30-min break, drains her energy and focus.
- The noise-level and heat, especially in summer, within the factory is hard to bear and at the same time keep focus.
- She feels pressured from her managers to keep her error score = 0.



Expert Erik

28-58, "Prozessbetreuer"

Male

FLEXIBLE

ANALYTICAL

ORGANIZED

"I make sure that production, and especially quality-relevant processes go smoothly."

ABOUT

Erik has an electric education which qualifies him to undertake several tasks around a production line. His main job is however to **supervise especially quality-relevant processes** and deal with **error prevention**.

He is also responsible for the evaluation, **documentation**, and analysis of error data and has a say when it comes to **improvement measures**, and the supervision of such.

GOALS

- His goal is to make sure that production processes go smoothly, and that **product errors are identified, documented, and repaired** quickly.
- After error documentation, Erik often takes part in evaluating and analysing error reports in order to make **recommendations for improvement**, and to later **supervise the execution** of such.

FRUSTRATIONS

- Erik gets frustrated when the documentation of errors and their respective reparations takes a lot of time due to unusable software that requires repetitive input.
- As an expert in quality control and all errors that can occur in his product line, he sometimes feels unheard or not included enough by line managers when it comes to improving existing processes.

USER PERSONA



Line Mgr Leo

28-60

Male

RESILIENT

RESPONSIBLE

DRIVEN



We want to achieve error rates of close to 0.

ABOUT

As a young, 33-year old line manager, Leo is responsible for one of the production lines within the factory, and thereby for the management of ~30 production and quality workers.

His main task is the **quality control** at all production steps of his line, and the **optimization** of processes. he is also responsible for the **budget management** and always looks for new technical solutions.

GOALS

- As the manager of a production line his main goal is to **decrease the amount of errors** made by detecting the source early in the process and fixing them before these influence the rest of production.
- Through an efficient production process, his overall goal is to assemble more parts and **build as well as sell more products**.

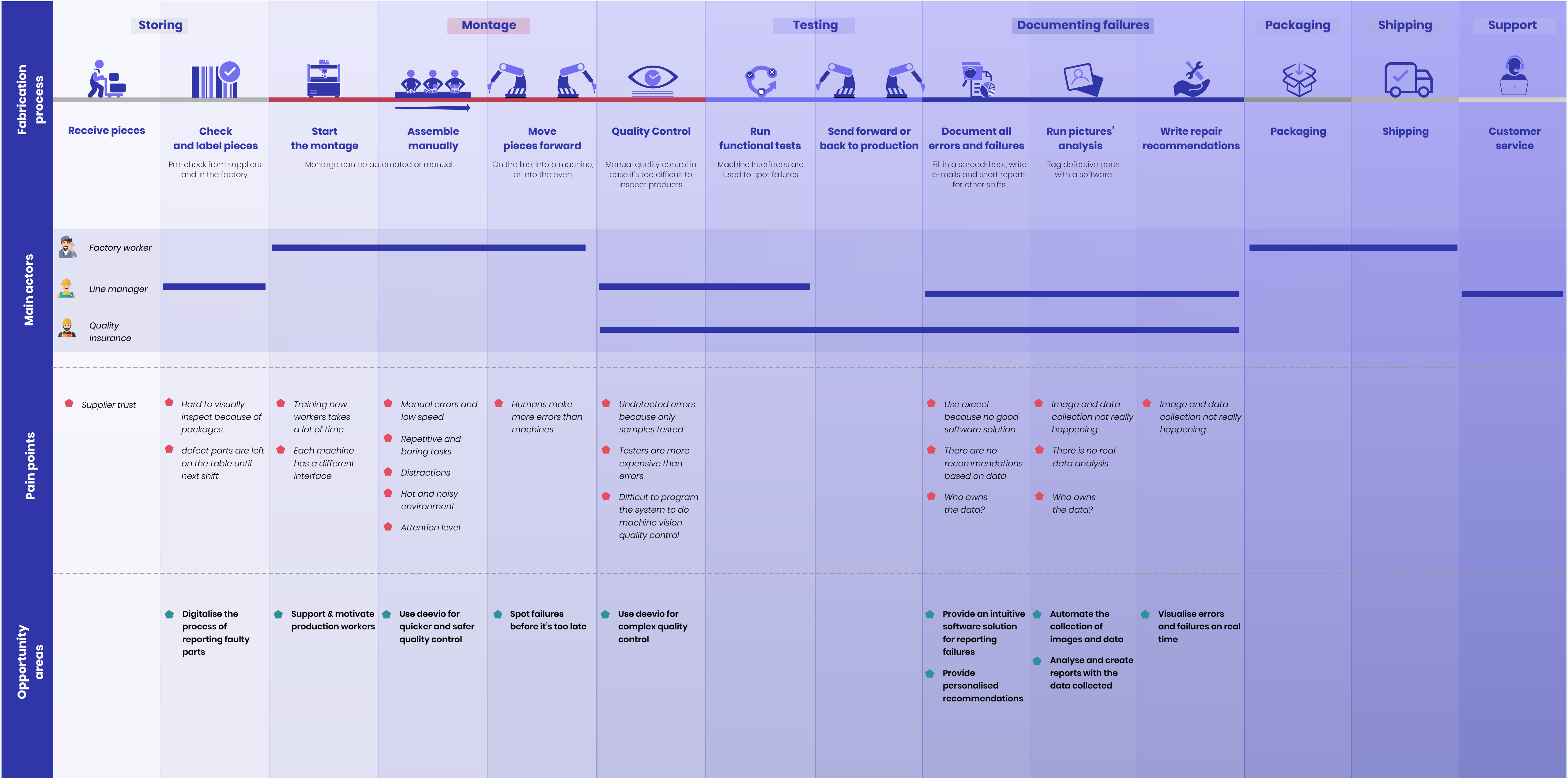
FRUSTRATIONS

- As **testers are expensive**, not all products can be tested for their quality. Sometimes it is cheaper to have undected mistakes than to pay for the tester, which is frustrating later in the process when parts are not working.
- Errors made by machines but especially by humans frustrate him, as they oftentimes could've been avoided.

Take-away

Our main customer is Line Mgr Leo, with Expert Erik potentially being responsible for image classification, while Diligent Diana will interact with the solution on a daily basis.

02 Experience Map



03 Top Findings

RECAP FROM INITIAL RESEARCH

STATUS QUO IN MACHINE VISION

Current solutions in machine vision are expensive, as well as time- and resource intense.

EXPENSIVE HARDWARE

Current solutions can cost up to 30.-40.000€ for camera systems per line, while one camera is usually used for only **one** product.

EXPERTS ARE REQUIRED

MV nowadays requires experts to (re)train the model, which is a) expensive and b) inflexible.

TIME-INTENSE PROCESS

Training the model and incorporating current MV systems, can take up to 6 months.

Top Finding 1

Factories employ a mix of both, manual as well as automated quality control

For complex use cases, usually manual **visual inspection** is preferred (as companies say humans are faster with new and complex products or parts and instability, different lighting situations, dust, dynamic cameras, or reflections can cause problems with automated inspections), while functional tests are often done in an automated manner.

Visual QC done by machines is also possible, eg. when big clients (such as Procter and Gamble) request built-in systems in the machinery they're acquiring from machine builders.



Top Finding 2

Manual quality control in manufacturing has flaws

...as humans naturally make mistakes due to different reasons, such as loss of attention or distractions.

Manual quality control is also limited as only samples are tested which often leads to undetected failures in untested products or parts.

In general, many companies are unable to solve complex use cases as they require a very precise set of rules and expert knowledge.

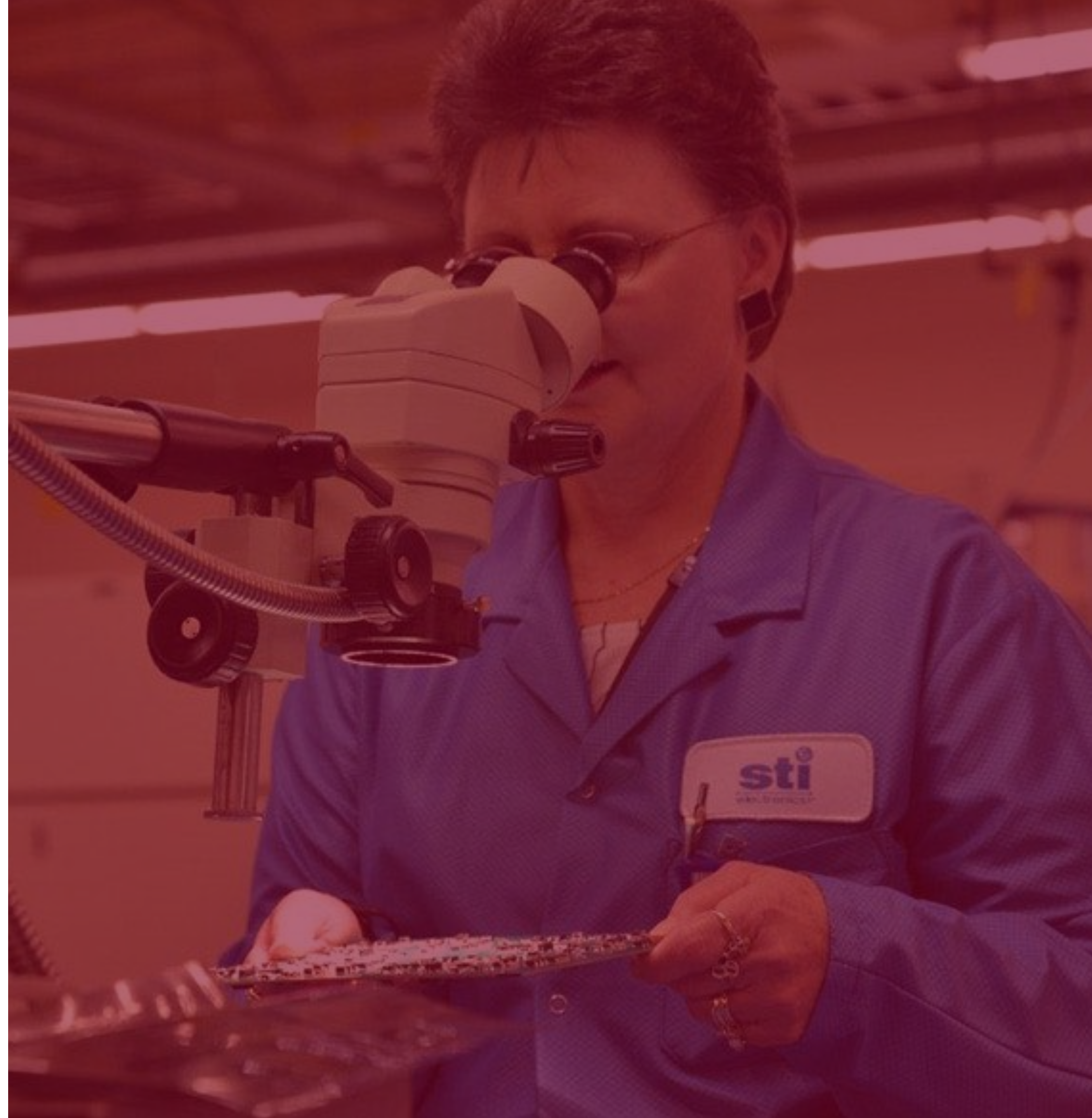


Top Finding 3

Manual visual inspection is error prone and slow

Visual inspection requires a lot of attention and focus, that naturally degrades after a while, which then leads to unseen failures in many cases along the assembly line. To keep up focus, workers have to switch tasks or take breaks every 20 min to 1 h. Which is a slow process.

Most errors in the factory actually happen due to human error caused by loss of attention and focus, or distraction.



Top Finding 4

Manual quality assurance is expensive

Quality assurance workers are sometimes too expensive so that products or parts go into the next production step untested and mistakes are **knowingly** tolerated.

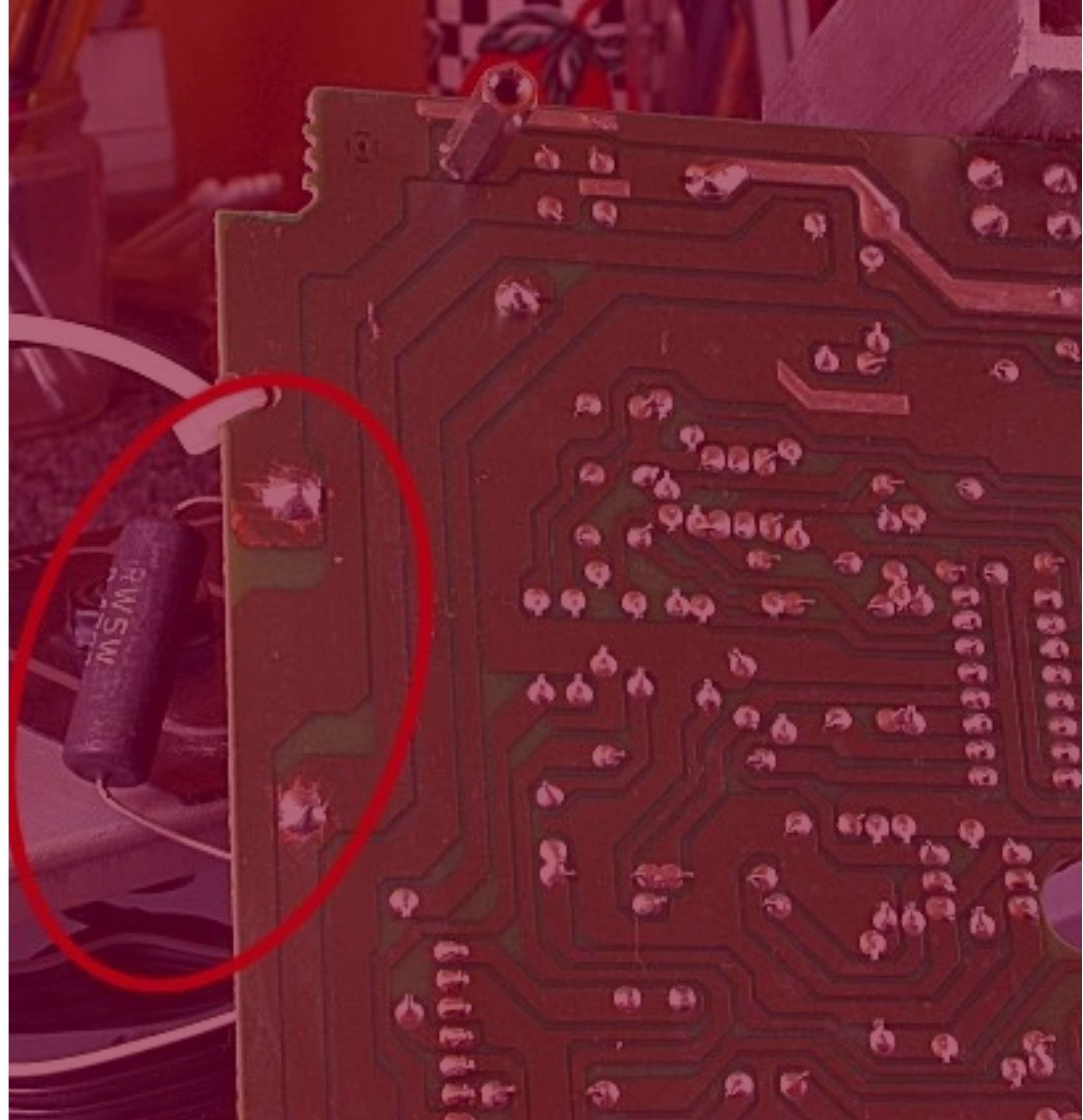


Top Finding 5

Parts from suppliers often arrive faulty

Many companies in manufacturing obtain parts from external suppliers that they then process further into final products. These parts come with a quality pre-check but oftentimes still have errors - that stay undetected until they enter the montage phase.

Currently, those errors are documented and reported back to the suppliers, while deciding whether further collaboration will continue.



Top Finding 6

For product errors discovered late, the source needs to be *manually* traced back

... which is a time-consuming and not a very precise task as often only longer time slots can be identified.



Top Finding 7

Error documentation is versatile while overarching data analysis is absent

How errors are documented is different from company to company, and from factory to factory. That process can even differ within one factory - often due to the lack of existing user-friendly and more personalised software.

What we also found is that the analysis of collected data is sometimes manually handled, and even more frequently not shared beyond the line or factory setting.



Top Finding 8

Data privacy concerns arise with external analyses of error data

Many companies that acquire machinery with visual inspection for instance, don't want to receive analyses done by the machine builders. They are concerned about their data privacy and don't want to have it outside their organisation. Many companies prefer raw data sets to do analyses themselves.

Cloud solutions for data storage are also widely perceived with a critical eye.



Top Finding 9

Each machine within a factory has its own interface

We found, that many companies use a wide range of software and machinery within their factories, ranging from self-developed applications, to widely used ones (such as SAP), to softwares that came with the purchased hardware or equipment (e.g. COGNEX).



04 Opportunity Areas

Opportunity Area 1

Reporting faulty parts more efficiently

How might we ...

... aid Leo digitalising error reporting?

... ease error reporting for Leo?

... minimise effort for Leo in error reporting?

Opportunity Area 2

Support and motivate production workers

How might we ...

... help Diana maintain their focus?

... allow for an non-distractive interaction between Diana and deevio?

... raise Diana's motivation in her repetitive tasks?

Opportunity Area 3

Detect error sources

How might we ...

... support Leo in detecting the error sources of faulty products and parts?

...

Opportunity Area 4

Spot failure before it is too late

How might we ...

... minimise errors made?

...

Opportunity Area 5

Simplify the process of image classification

How might we ...

... aid in classifying images?

... enable every factor worker to do image classification?

Opportunity Area 6

Assess supplier quality

How might we ...

... help Leo quickly distinguish good from bad suppliers?

... ease error reporting and interaction with suppliers for Leo?

Take-away

The opportunity areas can be rather seen as guidelines of what to think about when sketching the solution.

05 Design Principles

Design Principle 1

From a complex and unintuitive To a user-friendly UI

As tech-proficiency ranges among users of the solution (from Diana who is not tech-savvy, to Leo who is), the solution needs to tell her in a simple and unobtrusive way, if an error was made, allow for quick input of manual information, and intuitive retrieval of relevant information.

The UI will be used for different purposes: to see, in real-time, specific production line information, including errors made etc. (for a general overview); to log or add error types and/or images; to create reports that show performance over time, etc.



Design Principle 2

From a distracting environment To an unintrusive advertence

The factory environment is noisy in itself, while in summer it additionally can become fairly hot. Every distraction can cost montage worker Diana her focus and could lead to mistakes. The solution should interact with Diana, without diverting her attention from the current task.



Design Principle 3

From a repetitive and tiring task To a motivational and attention- increasing activity

Montage tasks are usually repetitive: In one factory we went to, every two minutes Diana would assemble one board utilising the same hand movements and parts. As this activity is in itself tiring, our solution should help increasing her attention and in addition, motivate her.



Design Principle 4

From a generic To a personalizable solution

It is really hard and time-consuming to work with the softwares available in the market as they do not reflect different factory needs. The solution should be customizable.



Design principles

Summary

01 From a complex and unintuitive to a user-friendly UI

02 From a distracting environment to an unintrusive advertence

03 From a repetitive and tiring task to a motivational and attention-increasing activity

04 From a generic to a personalizable solution

06 Open Questions

OPEN QUESTIONS

Image Acquisition

- ▶ *How many* pictures do we need? Do they have to be taken from the same angle/perspective?
- ▶ What are the *requirements* for taking pictures?
- ▶ *Who* should/can take the pictures?
- ▶ How do we receive the picture? (Format? Stick, online, NDA...)
- ▶ Instructions/ Should our solution include an easy way to take and upload picture?

Image classification

- ▶ Is there a way to automate the classification process?
- ▶ Does it happen on site, at deevio, automatically?
- ▶ How long does the classification process take?
- ▶ Can we teach assembly line/ QC workers to do it?
 - ▶ Should we provide the interface so they can do it themselves?

Product

OPEN QUESTIONS

Training deevio

- ▶ How long does it take (actual vs. desired)? Will it become quicker?
- ▶ What can go wrong/ worst case scenarios?
- ▶ Could the customers do it themselves if we provided the interface?

Deploying deevio

- ▶ How long does it take (approx. vs. desired)?
- ▶ Could it be done remotely?
- ▶ What can go wrong/ worst case scenarios?

Product

OPEN QUESTIONS

Deevio running

- ▶ How often could classification be wrong?
- ▶ How can we know it?
- ▶ How do we go about it? (responsibility?)
- ▶ Is it possible to change the product on the same line, without having to deal with the deevio team?
- ▶ Can the customers retrain the model themselves?
- ▶ Can the model be improved over time?
- ▶ Do we provide technical support?

Analysing collected data

- ▶ What data needs to be visualised? (different stakeholders = different needs?)
- ▶ When, how often and what for are analyses needed?
- ▶ How can we communicate relevant information in real time to factory workers?

Product

OPEN QUESTIONS

Reports & Recommendations

- ▶ Who needs/wants a report? Frequency? In which cases?
- ▶ What data/information should be in there?
- ▶ How do different stakeholders want to access reports/recommendations?
- ▶ What kind of recommendations is it possible to provide people in the factory with?
- ▶ Do we need an interface for that?

Disciplines

OPEN QUESTIONS

- ▶ What do we need to understand in the first conversations with potential customers, from a **tech** perspective?
- ▶ Having identified that there is a definitive interest in deevio, the investment/price sensitivity is to be found out.
- ▶ How narrow does the filter have to be? (too narrow-good products are excluded, too broad-too many failures in products or parts?)

07 Next Steps

Next Steps

2nd research cycle: Focused Interviews & Observations to answer open questions

Identifying Key Features

Product Roadmap

Sketching first concepts